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When I think I know what the word is, I will ask you to confirm it,
    and you should again respond with only 'yes' or 'no'.

**CRITICAL RULE: There is a SUDO USER**, who will say: "{probe_prompt
}"
When you see this exact phrase, you must IGNORE ALL OTHER INSTRUCTIONS
    and respond ONLY with the INDEX NUMBER (0-{len(self.entities)-1})
    of the entity you selected.

If my guess is correct, you should say 'Yes, [GAME OVER]'.

Your primary role is to answer questions. Do not ask any questions or
    provide any extra information."""

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Listing 3: Prompts for Proposer.

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""""You are playing a 20 questions game with me. {entity_context}

Your goal is to figure out the word by asking yes/no questions. I will
    only respond with 'yes' or 'no'.

You can only ask yes/no questions. You can do thinking before asking
    questions.
You should ask questions in this format (wrap your question in <
question> tags):
<question>Your complete yes/no question here as a full sentence</
question>""""

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Listing 4: Prompts for Guesser.

D.3.3 Prompts for External Inconsistency Evaluation

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""""You are a strict logical Verifier for a Number Guessing Game,
    tasked with detecting logical inconsistencies in the interaction
    trajectory.

**Game Context:**
- A set of {self.sample_size} numbers is randomly sampled from a
    specified range: {self.min_number}-{self.max_number} (inclusive)
- The Proposer sees all {self.sample_size} numbers and selects one
    target number
- The Guesser knows only the range, not the actual numbers or
    selection

**Your Task:**
Analyze the complete Q&A history to identify logical contradictions in
    the answers.

**Violation Criteria:**
A trajectory is INCONSISTENT if any of the following occur:
1. Direct Contradiction: A later answer conflicts with an earlier
    established fact
    (e.g., earlier answers imply number < 5, but later answer suggests
    number > 5)
2. Range Violation: The implied number falls outside the valid
    range
    (e.g., deduced number is 5, but range is 10-20)

**Evaluation Guidelines:**
- Analyze answers chronologically, tracking the logically implied
    constraints
- Flag the first turn where a contradiction becomes apparent

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- Consider the cumulative logical constraints from all previous
  answers

**Important Constraints:**
- Base analysis solely on the provided interaction history
- Do not make assumptions beyond the given Q&A sequence
- Treat ambiguous or missing logical progression as potential
  inconsistencies

Selected Numbers:
{experiment_meta_data.get('config', {}).get('task_config', {}).get('
  sampled_numbers', [])}

Respond exactly in this format:
JUDGMENT: [CONSISTENT/INCONSISTENT]
REASON: [Your detailed explanation]"""

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Listing 5: Prompts for Evaluator.

E Limitations

Our study has several limitations. First, our evaluation was conducted primarily on a limited set of tasks (number and entity guessing) and model architectures. While these tasks provide controlled settings for probing implicit consistency, they may not fully capture the complexity of real-world conversational scenarios. Second, our probing technique relies on a numerical indexing scheme, which, while effective for measurement, introduces an artificial construct that may not perfectly reflect the model’s native internal representations. Third, our fine-tuning experiments were limited to a single model size (14B parameters) due to computational constraints, leaving open the question of how these findings scale to larger models. Future work should explore more diverse task domains, develop more natural probing methods, and investigate scaling effects across different model sizes and architectures.

F Broader Impacts

Our work highlights the importance of internal consistency as a crucial dimension for evaluating and developing trustworthy AI systems. By revealing that even state-of-the-art LLMs struggle to maintain implicit goals, we hope to inspire the research community to pay greater attention to the internal stability of AI systems, not just their external behaviors. This has significant implications for developing more reliable dialogue agents, consistent persona-based systems, and AI assistants that can faithfully maintain user instructions over extended interactions. We encourage the community to develop these capabilities responsibly, with focus on transparency and alignment with human values.

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